Rent or Toxicity? Strategic Exploitability of Learning Market Makers with Public Wallet Information

Wargame Deliverable 2: Workshop 1 report

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6.S974 Games, Learning, and Security. Example 4: Algorithmic Market with Adaptive Agents

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a) Setting and research questions

The contested system. An *exchange* is a venue where buyers and sellers of an asset meet; Hyperliquid is one for crypto derivatives, and it is *on-chain*: its order matching runs on a public blockchain, so every order and trade is written to a ledger anyone can read, and every participant appears under a persistent public account address (a *wallet*). *Market makers* post standing offers to buy and to sell (a *quote*: a bid below and an ask above the current reference price, the *mid*; the gap is the *spread*; the set of standing orders is the *book*); *takers* accept them, which produces a trade (a *fill* for the maker). Prices live on a coarse grid (one step is a *tick*); standing orders at the same price are served first-in-first-out; and the exchange's own vault, with undisclosed strategies, supplies much of the standing liquidity. The public record of fills, with both wallets on each, is the *tape*. A maker's payoff on a fill is the difference between the price it got and the asset's value afterwards. Takers are *informed* (they know where the value is going and trade only when the maker will lose) or *uninformed* (they trade for other reasons). Being matched with the informed kind is *adverse selection*; flow with a high informed share is *toxic*. Three decisions matter: the makers' (how wide to quote, or, when the grid pins the quote at one tick, whether to keep an order standing); an *adversary*'s, a new maker we call the *entrant* (whether to *undercut*, i.e. quote a tick inside the makers' price and take their fills, or which fills to take); and the venue's (whether identities are public, how coarse the grid is, whether a place in the queue can be bought). The interaction is strategic in three ways. With enough competing makers, undercutting drives the spread down to the *competitive* level, where a maker's gains from uninformed takers just cover its losses to informed ones and it earns nothing on average. Independent learning makers settle on a wider spread than that without ever communicating (*tacit collusion*; such a set of makers is a *cartel*). What they earn above the competitive level is a *rent*: profit that exists only because they refrain from undercutting one another, and that a new maker could take by undercutting. But a wide spread is also the correct charge for adverse selection, and rent and toxicity look identical on the tape. An adversary must tell them apart, and its weapon, undercutting or taking only the benign fills, is what the makers' learned response to a rival's price (their *reaction function*) is tuned to punish. The venue's information policy decides what adversary and defenders are allowed to know.

Research questions and outcomes. RQ1 (value of identity): what is public wallet identity worth to an entrant beyond the anonymous tape, and through which mechanism? Outcome: the entrant's profit when it sees wallets minus its profit when it sees only the tape, in units of the cartel's rent (defined in §c), as a function of how long spells of toxic flow last and of how late the tape reveals a fill's outcome; on data, whether the makers who earn condition on identity. **RQ2 (exploitability as measurement):** how much of a learned cartel's rent can an adversary take or destroy, and does more rent mean more vulnerability? Outcomes: what the cartel loses when the adversary enters (its best-response loss), that loss as a fraction of the rent, and the adversary's own profit, each against how collusive the cartel was (its rent index); the exact best response; whether letting the cartel keep learning after entry defends it. **RQ3 (tick-constrained books and defences):** when the grid pins the quote, where does rent live, and which venue and maker levers change who holds the toxic flow? Outcomes: on Hyperliquid, each maker's realized profit per fill; in the simulator, the probability that an uninformed taker finds a standing order in a toxic spell, the venue's fee revenue, and for each lever the pair (defenders' profit alone, defenders' worst-case profit under attack).

Changes since the proposal, and feedback. The proposal's identity mechanism (a wallet's toxicity persists, so the wallet that just traded predicts the next) is worth nothing on the data: the same wallet trades again next only 2–5% of the time. We replaced it with the mechanism the data support: informed trading comes in spells (a hidden *regime*), and a wallet's fill reveals the spell before the price does, because a fill's outcome is only knowable once the price has moved (a *mark lag*). We also added a second mode of the same environment, the one-tick *queue game*, after the book data showed that the quoted spread on the major markets never leaves one tick. RQ2 became a measurement rather than a hypothesis, and its headline changed when we computed the exact best response. Following the professor's feedback we define every market term where it first appears, in the language of games and control; run one model in two modes; lead with results; and treat RQ1+RQ2 as the paper's spine, RQ3 as the project requires.

b) Literature and current practice

Algorithmic collusion. Calvano et al. [3] showed that independent Q-learners in a repeated pricing game converge

to supra-competitive prices sustained by learned punishment; Colliard, Foucault and Lovo [4] and Cont and Xiong [5] transplant this to market making in the Glosten–Milgrom [2] model (a dealer facing informed and uninformed takers), where the wide quote has a second reading as the price of adverse selection; Cartea, Chang and Penalva [6] show the tick grid itself sustains supra-competitive quotes; Dou, Goldstein and Ji [7] study AI traders colluding via price impact. We take the environment and learners from [4], the mechanism test from [3]. **Adversaries of learners.** Deng, Schneider and Sivan [8] and Brown and MacKay [9] study a strategic player facing no-regret or algorithmic opponents and *steering* their reaction functions; Gleave et al. [10] show a frozen reinforcement-learning policy is exploitable by driving it into states it never trained on; McMahan et al. [11] and Lanctot et al. [12] give the double-oracle (PSRO) loop, each side in turn best-responding to the other's current policies. These are the attacker's side of RQ2; our exact best response is computed, not learned, because a cartel whose learning has stopped is a known reaction function. **Identity.** Zhai [13] documents on Hyperliquid that public trader identity predicts adverse selection and returns; we turn that into an information set and price it. **Data.** Albers et al. [14] publish the Hyperliquid order-level archive we use alongside our own recorder.

Current practice (documented) vs. our assumptions. Documented by the venue [15] and the archive [14]: wallet addresses on all orders and fills; price-time priority; a five-significant-figure grid (one tick is 0.12 / 0.39 / 0.87 / 0.11 basis points, i.e. hundredths of a percent (bps), of the price on BTC/ETH/SOL/HYPE at current prices); per-address request budgets earned by filling; a priority mechanism for order placement; the protocol's own market-making vault (HLP) as a large standing liquidity provider whose strategy is not disclosed; liquidations routed to the book. How individual makers decide is not documented (the vault's strategy least of all); the practitioner template is Avellaneda–Stoikov [1]: quote around a fair value, skew for inventory, cancel on adverse signals. So what can be measured is the *outcome*, which fills each wallet gets and what they earn, and §d4 does that. Our modelling assumptions, all beyond documented practice: one unit traded per period at the best quote; takers of two types drawn from a wallet population with persistent types under a hidden two-state regime; the asset value redrawn each period (so benchmarks are closed-form); tabular Q-learning incumbents with the state variables of [4]; and, in the queue mode, a fixed one-tick spread with first-in-first-out service. **What we add:** exploitability of learned cartels as a measured quantity, up to the exact best response and the double oracle; identity as an information set between the anonymous tape and full knowledge, priced by regime persistence and the mark lag; the queue game with venue and maker levers on one profit–resilience frontier; and three data anchors: the one-tick book, who earns the spread, and identity-conditioned fill selection by the makers who earn it.

c) Game formulation

Agents and objectives. A *venue* (chooses the rules once; objective: fee revenue, or uninformed takers' access in stress); N *incumbents* (the defenders, the learning makers, $N \in \{2, 3\}$; each maximises its own discounted expected payoff, $\gamma = 0.95$); one *entrant* (the adversary; objective λ·own payoff $-(1-\lambda)$·incumbents' payoff, $\lambda = 1$ a profit-maximiser, 0 a damage-maximiser); *takers*, which are nature: a population of $M = 50$ wallets with types $\alpha_j \in [0, 1]$ (the probability that wallet j's order is informed), stratified quantiles of $\text{Beta}(\bar{\alpha}\kappa_\alpha, (1-\bar{\alpha})\kappa_\alpha)$ so the mean type is $\bar{\alpha}$ and κ_α sets the dispersion. Throughout, the *toxicity* of a flow is its informed share.

State and uncertainty. At period t: the asset's true value $V_t = m_t \pm d_t$, $d_t \sim \text{Exp}(\mu)$, redrawn each period around the public mid m_t; a hidden regime $z_t \in \{\text{calm}, \text{toxic}\}$ kept with probability ρ_z, else redrawn from a law with stationary mean toxicity $\bar{\alpha}$; last period's quotes h_{t-1}; in the queue mode, the queue of standing orders. Which wallet trades depends on the regime (toxic: wallet j arrives with weight $\propto \alpha_j$; calm: $\propto 1 - \alpha_j$), so at $\kappa_\alpha = 1$ the two states have toxicity 0.15 and 0.65 around $\bar{\alpha} = 0.3$: *who just traded is evidence about the state the next trader shares.*

Actions. Spread mode: each maker posts a *half-spread* $h \in \{1, \ldots, K\}$ ($K = 24$, tick $\theta = 0.5$), i.e. offers to sell at $m_t + h\theta$ and buy at $m_t - h\theta$; the lowest h on the book is the *best quote* (the *touch*). Queue mode: the spread is pinned at h and each maker chooses REST (keep a standing order at the touch), OUT, or REST-AND-PAY (pay the venue's priority fee f per period and be served ahead of non-payers). Venue levers: identity public or not, the tick, the fee. Resource constraints: one unit per fill, an inventory cap (never binding), and the venue's request budget (so far modelled only through the fee).

Information. Incumbents are tabular Q-learners whose state is their own last h and the best rival's last h; they are identity-blind by construction. The entrant's information set is a treatment: $\mathcal{F}^{\text{anon}}$ (the public tape: for every fill its direction, its price, and, L periods later, its *mark*, the value V_t against which the fill's profit is scored) $\subset \mathcal{F}^{\text{id}}$ (plus the wallet on every fill) $\subset \mathcal{F}^{\text{oracle}}$ (plus the regime). On $\mathcal{F}^{\text{id}}$ the entrant runs a Bayesian filter (posteriors over wallet types from scored fills, a belief over the regime from the sequence of wallets) and acts on α_{next}, its implied probability that the next taker is informed; on $\mathcal{F}^{\text{anon}}$ the same filter sees only outcomes, which arrive L periods late.

Sequence of play (one period). (1) Makers commit actions simultaneously. (2) Nature draws the regime, then a wallet, then whether its order is informed. (3) The taker hits the best quote: an informed taker trades iff the quote is strictly inside V_t (so only when the maker loses); an uninformed one trades iff a private valuation $m_t + \mathcal{N}(0, \sigma_L^2)$ crosses it. One unit goes to a maker at the best quote (random tie split in spread mode; the front of the queue in queue mode, which then rejoins at the back). (4) The fill (direction, price, wallet) becomes public; its mark V_t becomes public at $t + L$. (5) Payoffs: the filled maker earns $\pm(\text{price} - V_t)$, minus the fee if paying; the venue collects fees. Horizon: $T = 100$

periods per episode; episodes are a training device, the wallet population and regime chain persist across them. Training: 2×10^4 episodes; evaluation: 300–500 episodes with the incumbents *frozen* (no further updates, greedy play) and the same random draws across treatments (common random numbers).

Payoffs and benchmarks (closed form). The market-wide expected one-period payoff when the best quote is h and the informed share is α is $\Pi(h;\alpha) = (1-\alpha)\,2x\bar{\Phi}(x/\sigma_L) - \alpha\mu e^{-x/\mu}$, $x = h\theta$, with $\bar{\Phi}$ the standard normal tail probability ($\sigma_L = 8$, $\mu = 5$); the makers at the best quote split it. $h^C = \min\{h : \Pi \geq 0\}$, the *break-even* quote, is the Nash point of the one-shot game (any $h > h^C$ is profitably undercut); $h^M = \arg\max_h \Pi$ is the joint monopoly; $(h^C, h^M) = (2, 13), (4, 14), (8, 17)$ and $\Pi^M - \Pi^C = 1.90, 1.38, 0.66$ at $\bar{\alpha} = 0.1, 0.3, 0.5$. Let Π_{nominal} be the frozen incumbents' expected payoff per period when they play alone. The *rent index* $C = (\Pi_{\text{nominal}} - \Pi^C)/(\Pi^M - \Pi^C)$ places them between Nash (0) and monopoly (1). With adversary A present they earn $J(A)$; $L^A_{\text{BR}} = \Pi_{\text{nominal}} - J(A)$ is their *best-response loss*, $\kappa_A = L_{\text{BR}}/(\Pi_{\text{nominal}} - \Pi^C)$ the *fraction of rent destroyed*, and *rent units* divide any payoff by $\Pi^M - \Pi^C$. The game is general-sum: the defenders' loss, not the adversary's gain, is the statistic. *Small example* ($\bar{\alpha} = 0.3$): two incumbents at $h = 14$ split $\Pi(14) = 1.50$; an entrant at 13 takes every fill and earns $\Pi(13) = 1.49$, the incumbents 0; if they answer at 12, the entrant at 11 earns 1.39; the chase ends at $h^C = 4$, $\Pi = 0.12$. A learned cartel is a reaction function that does *not* chase every time. *Why this structure:* one unit per period and an i.i.d. value make rent measurable; the grid and queue are the venue's rules; tabular Q-learners are the literature's incumbents; the regime and mark lag are what the data forced on us.

d) Preliminary results

d1. Learners coordinate above the Nash point. Two or three Q-learners settle at 10.8, 14.9, 17.4 ticks at $\bar{\alpha} = 0.1, 0.3, 0.5$ against Nash $2, 4, 8$ and monopoly $13, 14, 17$: $C = 0.83, 0.86, 0.78$ (seed sd 0.01–0.02), replicating [4]; a grid over $N \in \{2, 3, 5\}$, learning rate and initialisation gives $C = 0.71$–0.87 throughout. The mechanism is *restraint*: in 55–83% of visited states a rival sits at or inside the monopoly quote, so undercutting it would pay this period, and in 96–100% of those the learner's greedy action forgoes the undercut. Fills rotate (no leader); a forced undercut is answered in kind and the cartel re-forms within a few periods.

d2. Exploitability. *Rule adversaries.* Four fixed entrant policies: the *competitive quoter* (always quotes h^C), the *undercutter* (one tick inside the best incumbent quote), the *markout entrant* (estimates the toxicity α from the scored outcomes of its own fills and undercuts only while the incumbents' quote is above the break-even for that estimate), and the *oracle* (the same rule handed the true $\bar{\alpha}$). Over 180 frozen cartels ($C \in [0.42, 0.97]$, 500 episodes each) the proposal's hypothesis (more collusive, more exploitable) holds only by construction (the competitive quoter pushes every cartel below Π^C, so L_{BR} has slope one in C); the substantive version fails: κ_A is 0.71–0.73 for the other three, flat in C. The markout entrant keeps 79–96% of the monopoly rent, and keeps it when the incumbents resume learning: the cartel re-forms around it. *The exact best response* (Figure 1, 60 cartels). A frozen cartel is a deterministic reaction function with known transitions and closed-form rewards, so the entrant's problem is a finite-horizon Markov decision process that backward induction solves exactly for any objective $\lambda \cdot$ (own profit) $- (1-\lambda) \cdot$ (incumbents' profit); the computed value and the simulated one agree to 0.02. It takes everything: $L_{\text{BR}} > C$ in 60 of 60 cartels, the entrant earns 1.08–1.20 rent units, which is the monopoly profit itself, and $\kappa_A = 1.14$–1.29; the rules reach 55–70% of it. How: it undercuts the *reaction*, not the last fill (one tick inside what the incumbents are about to quote, 100% of periods), and holds the cartel in states it never visits when playing alone (86–99% of periods), where its learned response sends it *wider*: $10.8 \to 15.9$ ticks at $\bar{\alpha} = 0.1$, $17.8 \to 21.9$ at 0.5. The damage-maximiser ($\lambda = 0$) steers the cartel *tight*, to 4–7 ticks, inside its break-even, by quoting above it, takes no fills, earns nothing, and leaves the incumbents absorbing toxic flow at losing quotes: $L_{\text{BR}} = 1.3$ at $\bar{\alpha} = 0.3$, 2.7–2.8 at 0.5, two to three times the rent, at no cost. *Double oracle:* the cartel resumes learning for 3000 episodes against the fixed best response, is frozen again, and the entrant re-solves; three rounds. Re-learning buys back half of what the now-stale attacker takes (its profit $1.2 \to 0.5$) and nothing against the re-solved one ($L_{\text{BR}} \geq 0.9C$ in every cartel, every round), while the cartel's own rent erodes $0.83 \to 0.69$. *Robustness:* cartels keep $C = 0.80$–0.86 and $\kappa_A = 0.65$–0.75 (rules) on any market calmer than their training one; on a more toxic one their rent collapses ($C = 0.23$ for the 0.1-cartel at 0.5); adversaries held out when choosing a defence preserve every ordering (rank correlation 0.75–0.96).

d3. What identity is worth in the simulator (RQ1). Call the entrant on $\mathcal{F}^{\text{id}}$ the *identity entrant* and the one on $\mathcal{F}^{\text{anon}}$ the *anonymous entrant*; identity's value is the first's profit minus the second's, paired by seed. Under the proposal's mechanism (the wallet that just traded trades again next with probability ρ) identity is worth +0.15 to +0.30 rent units at $\rho = 0.9$–0.99 and exactly zero at $\rho = 0$, which is where Hyperliquid is. Under the regime mechanism with the mark revealed immediately ($L = 1$) identity adds only +0.03 to +0.06: the last fill's *outcome* is as good a regime signal as its author. With the lag, the anonymous entrant's profit falls as the mark arrives later while the identity entrant's does not, so the increment grows from +0.06 to +0.16 (± 0.07) at $\bar{\alpha} = 0.5$, $\rho_z = 0.99$, and peaks at $L = 20$ for $\rho_z = 0.9$, where a regime lives about ten arrivals: identity's value grows with the lag up to the regime's lifetime, and is zero where even the toxic state leaves an undercut profitable ($\bar{\alpha} \leq 0.3$ against 11-tick cartels). At $\kappa_\alpha = 5$ it can hurt: in three of ten cartels the incumbents' reaction punishes the identity entrant's moving quote; the reaction function is part of the game.

d4. Hyperliquid data (Table 1, Figure 2). Four sessions of our own recording of the best bid and ask over two

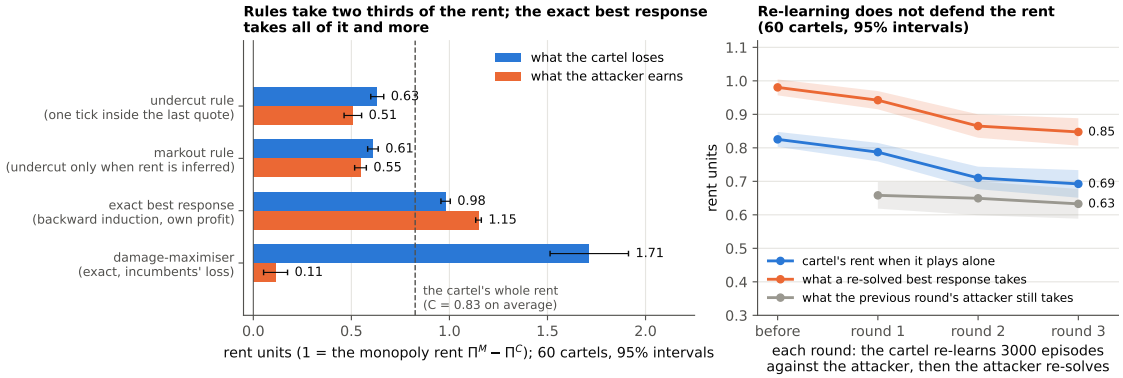


Figure 1: Left: what each adversary takes from a frozen cartel (the cartel's loss and the adversary's own profit, in rent units; means over 60 cartels with 95% intervals); the dashed line is the cartel's entire rent. Right: the double oracle, by round: the cartel's rent alone, what the re-solved best response takes, and what the previous round's attacker still takes.

	BTC	ETH	SOL	HYPE
taker's cost vs the mid (effective spread): fills at the touch / walked fills / per order (bps)	0.12 / 2.35 / 0.27	0.42 / 3.20 / 0.60	0.92 / 5.20 / 1.09	0.17 / 3.57 / 0.40
10-s maker markout, at the touch / walked (bps)	−0.41 / −0.80	−0.56 / −2.26	−0.27 / +0.98	−0.45 / −0.01
next fill's markout after a toxic vs a benign wallet, given what an anonymous quoter sees (bps, 327 h)	−0.64	−1.01	−0.59	−0.59
10-s markout at the touch: top-10 makers / everyone else (bps)	−0.05 / −0.36	+0.22 / −0.68	+0.53 / −0.50	+0.01 / −0.56
per-maker rank correlation, first session vs later (n makers)	0.80 (113)	0.53 (80)	0.70 (53)	0.59 (70)
top-10 share of touch fills: benign / unseen / toxic wallets	.11 / .13 / .03	.43 / .34 / .06	.42 / .37 / .09	.32 / .26 / .12

Table 1: Hyperliquid, Oct. 2026, four book sessions (one tick = 0.12 / 0.39 / 0.86 / 0.11 bps). Standard errors cluster by five-minute block; every top-10-vs-rest gap exceeds four SEs. "What an anonymous quoter sees": whether the price had already moved against the previous maker. Wallet types from the first night, scored on the later sessions.

weeks (25 h) and 449 h of fills with both wallets; 490k fills are *scored*: a maker's *markout* at τ seconds is its profit on the fill against the mid τ seconds later, in bps of the price (the data counterpart of the simulator's payoff). *The price cannot move:* the quoted spread on BTC/ETH/SOL is one tick essentially always (its 90th percentile equals its median). A fill at the touch pays exactly that tick and then loses 0.2–0.4 bps to the mid within ten seconds on every coin; the taker's cost is in *depth*: one order in ten is larger than what stands at the best price, takes successive worse prices (*walks* the book) and pays twenty ticks, and on the deep books those are the informed orders. *Someone still earns the spread:* classifying wallets as market makers by their activity (mostly standing orders, both sides) over the full window, the ten most active of them break even or earn where everyone else loses; at the same price they earn 0.3–1.0 bps per fill more, and their ranking persists from the first session to the later ones (rank correlation 0.53–0.80, $p < 10^{-6}$). *How: identity, not the tape.* They do not leave the touch after a large recent price move at any timescale (30 s to 5 min, or the mid's move in the 10 s before each fill: correlations −0.11 to +0.13). But classify every taker wallet on the first night by the markouts it inflicted (benign / toxic thirds) and score the three later sessions: the top makers take 0.03 / 0.06 / 0.09 / 0.12 of the toxic wallets' fills against 0.11 / 0.43 / 0.42 / 0.32 of the benign ones (BTC/ETH/SOL/HYPE; 2.6–6.8×, $p < 10^{-20}$), the same gap whether the tape was calm or stressed, and still 1.4–4.1× on fills after which the mid did not move within a second, i.e. not *pick-offs* (stale quotes hit just before a price move, which speed alone would avoid). The classification holds (toxic wallets cost 1.1–2.1 bps more in the later sessions). Identity-conditioned fill selection operates on the real book: RQ1's mechanism, observed.

d5. When the price cannot move: the queue game and the levers (RQ3). With the spread pinned at the break-even h^C a fill is worth +0.15 on average, +1.10 in the calm state and −2.26 in the toxic one: all value is in *selection*, choosing which fills to take by when to rest. The entrant rests iff its forecast of the next taker's toxicity makes the fill worth taking, from $\mathcal{F}^{\text{id}}$ or from $\mathcal{F}^{\text{anon}}$. Against incumbents that always rest, the identity entrant earns 0.14–0.16 per period (its fills 16% informed vs the incumbents' 29–31%) and its edge over the anonymous one grows with the lag (+0.04 → +0.10). Incumbents that learn need a *fringe* of always-resting makers (else a toxic spell empties the book for good), and tabular Q-learning cannot learn to queue (time priority pays only after consecutive periods in line, which exploration breaks); incumbents running a no-regret rule (EXP3) over a menu of complete policies (always rest, never rest, select on the tape) learn to select within a few hundred episodes, and identity is then worth *more*, +0.05 to +0.06 at every lag, five of five seeds, while the fringe's payoff falls from ≈ 0 to −0.02 to −0.10: identity-blind resting liquidity is the counterparty of the selection rent, in the simulator as on the book. *Venue levers.* Under anonymity with a late mark ($L \geq 5$) the incumbents cannot tell a toxic spell in time and keep resting (an uninformed taker is filled 71–75% of the time in the toxic state; the incumbents earn ≈ 0); with identities they withdraw (fill probability 0.23–0.30; incumbents earn 0.12–0.14) and an identity-blind maker among them earns −0.13 to −0.29: anonymity is a transfer from makers to uninformed takers in stress, and neither setting defends the rent. The priority fee turns the queue into an all-pay auction for position: at the venue's revenue-maximising $f = 0.5$ it collects two to four times the makers' total rent and leaves every maker at or below zero; a cheap fee doubles the best-informed maker's profit; the fringe pays. *Makers' levers* (20

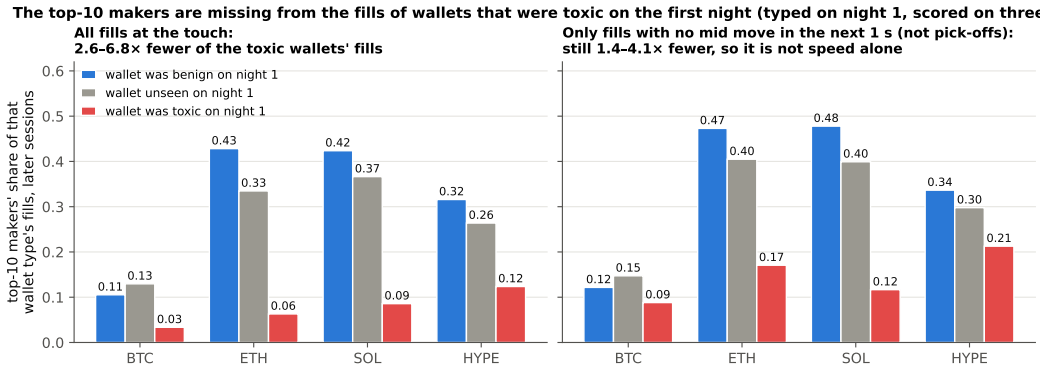


Figure 2: Identity selection on the book: the top-10 makers' share of fills at the touch in the three later sessions, by the taker wallet's type on the first night ("unseen": fewer than 30 fills then). Left: all touch fills; right: only fills after which the mid did not move within 1 s, which speed alone cannot explain.

cartels wrapped in each defence, seven adversaries including a Q-learning one): requoting only every 2–5 periods costs everything ($\kappa_A \to 1.0$–1.3); randomising the quote costs profit and buys nothing; never quoting below the break-even of the identity-based toxicity forecast is the only lever that raises both profit alone and worst-case profit under attack, and only where the toxic regime bites ($\bar{\alpha} = 0.5$: worst case $0.08 \to 0.19$); it still leaves the rule adversary 75–80% of its take. Nothing in the game defends the rent; the levers decide who holds the toxic flow.

Limitations. The exact best response uses the cartel's reaction function (oracle knowledge of the *policy*); it is the true upper bound; the attacker that identifies it from public quotes is not yet built (§f). Incumbents are tabular and frozen at evaluation; the double oracle is three rounds. Data: four sessions over two weeks, four pairs; wallet types need ≥ 30 fills, so only active takers are typed; a faster price feed from another exchange is not excluded as the top makers' signal on pick-offs (it cannot produce the gap on fills that were not pick-offs). Depth, where the data put the taker's cost, is not modelled.

e) Prototype

State. A Python package (`github.com/pradeepu191/wargame`; 59 tests): `sim/` (environment with wallets, regime and mark lag; Q-learners; rule adversaries and the entrant's Bayesian filter on $\mathcal{F}^{\text{anon}}/\mathcal{F}^{\text{id}}/\mathcal{F}^{\text{oracle}}$; the queue mode with fee and EXP3 incumbents; closed-form benchmarks), `experiments/` (training, entry, exploitability, identity grids, queue game, defences, exact best response, double oracle), `analysis/` (recorder, markouts, per-maker scoring, identity selection), cluster scripts, and `results/REPLICATION_NOTES.md`, a log of every result with its command. **How it operates.** `python experiments/demo.py` runs a small instance in seconds: it prints the benchmarks ($h^C = 4$, $h^M = 14$), plays the shipped two-maker cartel alone ($C = 0.86$), enters three adversaries under common random numbers (cartel loses / entrant earns: undercut 0.41/0.33, markout 0.47/0.41, exact best response 0.94/1.09) and traces the best response:

```
t  inc0 inc1 entrant  taker         filled  payoff
0   15   24      12  noise    buy  entrant  +1.21
5   21   19      14  informed buy  entrant  -5.77
```

The entrant sits one tick inside what the incumbents are about to quote and holds them where they never trained. `-train` fits a fresh cartel first (two minutes; $C = 0.83$, best response takes 0.91). The implementation is §c line by line: the period's sequence is `env.step`, an information set is what an agent's `act` may read, the benchmarks the closed form above. **Remaining:** the public-information attacker (§f); learned entrants and re-learning cartels in one PSRO loop; depth as an action; the request budget; a replay harness scoring a policy on recorded fills.

f) Plan for Workshop 2

Modelling and implementation. (1) *The attacker with public information only*: an entrant on $\mathcal{F}^{\text{anon}}$ that identifies each incumbent's reaction function from the tape (every answer to (own, rival) is public; the entrant sets "rival" with its own quote) and solves it; report its gap to the exact best response against the probing budget. (2) *Both sides learning*: PSRO with re-learning cartels and learned entrants: the equilibrium's rent and κ_A, and whether adversarially trained cartels are less exploitable or just less collusive. (3) *Queue-mode attacks*: steering no-regret incumbents; draining the request budget. (4) *Resilient designs*: venue levers (anonymity, tick, fee, request allowance) and maker levers, each a point (profit alone, profit in the worst 5% of episodes under attack) against all three attackers, plus uninformed takers' fill probability in stress, so "robust" cannot mean "withdrew"; paired intervals throughout. **Data.** An always-on recorder, now running (weeks: the identity-selection gap by coin, hour, regime); the Oct. 2025 liquidation cascade as the stress case; the Dec. 2025 order-level archive for which action carries the selection; preprint mid-December.

References

[1] M. Avellaneda, S. Stoikov. High-frequency trading in a limit order book. *Quantitative Finance* 8(3), 2008.

[2] L. R. Glosten, P. R. Milgrom. Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *J. Financial Economics* 14, 1985.

[3] E. Calvano, G. Calzolari, V. Denicolò, S. Pastorello. Artificial Intelligence, Algorithmic Pricing, and Collusion. *AER* 110(10), 2020.

[4] J.-E. Colliard, T. Foucault, S. Lovo. Algorithmic Pricing and Liquidity in Securities Markets. *Review of Financial Studies*, 2026.

[5] R. Cont, W. Xiong. Dynamics of market making algorithms in dealer markets: learning and tacit collusion. *Mathematical Finance* 34(2), 2024.

[6] Á. Cartea, P. Chang, J. Penalva. Algorithms and Supracompetitive Prices in Electronic Markets: The Impact of Tick Size. SSRN 4105954, 2026.

[7] W. W. Dou, I. Goldstein, Y. Ji. AI-Powered Trading, Algorithmic Collusion, and Price Efficiency. NBER WP 34054, 2025.

[8] Y. Deng, J. Schneider, B. Sivan. Strategizing against No-Regret Learners. *NeurIPS* 2019.

[9] Z. Y. Brown, A. MacKay. Competition in Pricing Algorithms. *AEJ: Microeconomics* 15(2), 2023.

[10] A. Gleave, M. Dennis, C. Wild, N. Kant, S. Levine, S. Russell. Adversarial Policies: Attacking Deep Reinforcement Learning. *ICLR* 2020.

[11] H. B. McMahan, G. J. Gordon, A. Blum. Planning in the Presence of Cost Functions Controlled by an Adversary. *ICML* 2003.

[12] M. Lanctot et al. A Unified Game-Theoretic Approach to Multiagent Reinforcement Learning. *NeurIPS* 2017.

[13] D. Zhai. Public Trader Identity: Adverse Selection and Return Predictability. arXiv:2608.04373, 2026.

[14] J. Albers, M. Cucuringu, S. Howison, A. Y. Shestopaloff. An Open Book: Level 4 Order Book Data from the Hyperliquid Exchange. SSRN 6465720, 2026.

[15] Hyperliquid Documentation. `https://hyperliquid.gitbook.io/hyperliquid-docs`, accessed Oct. 2026.